

```

MATCH (t:Terminal)

CALL {

WITH t

MATCH (t)<-[rPrev:MADE_TRANSACTION]-()
WHERE datetime(replace(rPrev.tx_datetime,' ','T')).year = 2018
AND datetime(replace(rPrev.tx_datetime,' ','T')).quarter = 4

WITH rPrev.amount AS amount
ORDER BY amount

WITH collect(amount) AS amounts

WITH amounts, size(amounts) AS n

RETURN
CASE
WHEN n % 2 = 1
THEN amounts[toInteger(n/2)]
ELSE (amounts[toInteger(n/2)-1] + amounts[toInteger(n/2)]) / 2.0
END AS medianQ4
}

MATCH (t)<-[rCurrent:MADE_TRANSACTION]-()

```

```

WHERE datetime(replace(rCurrent.tx_datetime,' ','T')).year = 2019
AND datetime(replace(rCurrent.tx_datetime,' ','T')).quarter = 1
AND rCurrent.amount > medianQ4 * 1.3

RETURN
t.id AS terminalId,
rCurrent.tx_id AS transactionId,
rCurrent.amount AS amount,
medianQ4,
medianQ4 * 1.3 AS threshold
ORDER BY amount DESC
LIMIT 100;

```

With outliers

```

MATCH (t:Terminal)
CALL {
WITH t
MATCH (t)<-[rPrev:MADE_TRANSACTION]-()
WHERE datetime(replace(rPrev.tx_datetime,' ','T')).year = 2018
AND datetime(replace(rPrev.tx_datetime,' ','T')).quarter = 4

```

```

WITH rPrev.amount AS amount

ORDER BY amount

WITH collect(amount) AS amounts
WITH amounts, size(amounts) AS n

RETURN

CASE
WHEN n % 2 = 1
THEN amounts[toInteger(n/2)]
ELSE (amounts[toInteger(n/2)-1] + amounts[toInteger(n/2)]) / 2.0
END AS medianQ4
}

MATCH (t)<-[rCurrent:MADE_TRANSACTION]-()
WHERE datetime(replace(rCurrent.tx_datetime,' ','T')).year = 2019
AND datetime(replace(rCurrent.tx_datetime,' ','T')).quarter = 1
AND rCurrent.amount > medianQ4 * 1.3

SET rCurrent.isOutlier = true

RETURN count(rCurrent);

```